CHANDAK CHAKMA

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RESEARCH INTERESTS

- **Efficient generative world models:** fast video world models that sample many possible futures in one pass. Research on whether the **shape** of these sampled futures matches the real spread of outcomes, and how to fix it.
- **RL for LLM reasoning (RLVR):** reward design beyond ground-truth labels, label-free supervision, and the pass@1 vs. pass@k trade-off.

EDUCATION

University of Dhaka — **Dhaka, Bangladesh**
B.Sc. in Robotics and Mechatronics Engineering — Sep 2022 – Oct 2026

PUBLICATIONS

1. **C. Chakma**, S. N. Sakib, N. Haque, S. E. Arman. "Unlearnable, or Unmeasured? On the Reliability of Difficulty Labels in RLVR." **Accepted**, Transitioning from Pre-Training to Post-Training (**Pre→Post**) Workshop, **NeurIPS 2026**. arXiv:2609.40115.
2. M. H. Promit, **C. Chakma**, M. J. A. Sourov, S. Rahman. "Escape Without a Recipe: Maneuver-Agnostic Mars Rover Recovery from Granular Entrapment." **Accepted (Poster)**, **IEEE IROS 2026 Workshop** (SRW), Pittsburgh.

RESEARCH EXPERIENCE

Cortex AI Lab, University of Dhaka — **Dhaka, Bangladesh**
Research Assistant — Aug 2026 – Present

- **Efficient generative world models (ongoing):** studying whether one-pass video world models (DeltaWorld) predict a correct distribution of futures, not only good single guesses; building a sampling-based evaluation on Kinetics, VSPW, and KITTI.
- **RLVR reproducibility study (first author, NeurIPS 2026 workshop, accepted):** showed that "unlearnable prompt" sets reported in recent RLVR work depend heavily on measurement choices — the same data yields 74–228 prompts under different aggregation rules, and much of the cross-seed disagreement is reproduced by re-evaluating the same checkpoint. The slow-learning effect itself survives: these prompts learn at about one third of the normal rate.
- **Theory:** derived a binomial sampling model for how many rollouts a reproducible difficulty label needs (8–16 rollouts give only 0.63–0.73 agreement; 0.90 needs about 242), turning an informal practice into a stated sample-size requirement.
- **Mechanism check:** showed that part of the reported gradient-similarity gap comes from hard prompts having fewer correct rollouts; matching the sample count cuts the easy-to-unlearnable ratio from 2.33× to 1.54×.
- **Experimentation:** trained and evaluated Qwen2.5-0.5B with GRPO across five random seeds on a single consumer GPU; built the seed-sweep, logging, and evaluation harness end to end.

University of Dhaka – Robotics Lab — **Dhaka, Bangladesh**
Research Student, advised by Dr. Sejuti Rahman — April 2025 – Present

- **Undergraduate thesis (IROS 2026 workshop, accepted as poster):** trained a maneuver-agnostic recovery policy for a wheeled rover trapped in granular regolith, using a grain-level material point method (MPM) simulator. Across five training seeds the policy escapes 62.5% of verified traps on average (best seed 26/32), against 7/32 for the best of ten scripted controllers.
- Built simulator-integrity checks; an audit caught a coupling-buffer fault that silently dropped 9.7% of contact nodes.

PROFESSIONAL EXPERIENCE

NoeticX (Contract) — **Remote, USA**
Assistant AI Engineer — Oct 2025 – April 2026

- Built a **Bangladeshi NID Card OCR API** (PaddleOCR, OpenCV, FastAPI) returning name, date of birth, and NID number as structured JSON.
- Developed a **Tablet Strip OCR** pipeline reading drug data from medicine packaging under real-world image noise.
- Created a **Natural Language CSV Data Analyst** (PandasAI, Plotly) for querying and plotting CSV data in plain language.

ENGINEERING PROJECTS

Production RAG + ReAct Agent | GitHub | Python, LangChain, FastAPI, Docker

- Hybrid retrieval (dense + BM25, Reciprocal Rank Fusion), cross-encoder reranking, OpenTelemetry tracing, and RAGAS evaluation.

Autonomous Research Agent | GitHub | LangGraph, GPT-4o, Pinecone, Python

- Literature-search agent that searches the web, stores knowledge in Pinecone, and writes structured reports, with **human-in-the-loop** approval.

LLM Hallucination Detection | Python, NLP, LLMs

- Detects hallucinations by splitting LLM outputs into claims and checking each claim against source documents.

TECHNICAL SKILLS

Languages: Python, C/C++ **ML / DL:** PyTorch, TensorFlow, Hugging Face Transformers, NumPy, Matplotlib
LLM Post-Training: RLVR (GRPO), SFT, LoRA, QLoRA **LLM Apps & Agents:** RAG, LangChain, LangGraph, Pinecone
World Models & Vision: video world models, OpenCV, PaddleOCR **Engineering:** FastAPI, Docker, Git/GitHub
Others: Data Structures & Algorithms, Competitive Programming (3+ years), Probability & Statistics

SELECTED AWARDS

2nd Runner-up among 140 teams | Datathon Bitfest — Machine Learning Competition, KUET — 2025

28th Place | Battle of Brains Programming Contest, University of Dhaka (as a first-year undergraduate) — 2023